



Logistics Data Management

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Material Master Data Structures

Material Master Data structure

Some material master datas are valid everywhere, but some only on respective organisational level. For instance, a material may be shipped differently in different plants. The different datas on the respective organisational levels are stored in different database tables, approximately following the principle of relational database management. The data structures are:

Table	Key	Description
MARA	MATNR: Material Number	General Material Data
MAKT	MATNR: Material Number SPRAS: Language	Descriptions
MARC	MATNR: Material Number WERKS: Plant	Plant Data
MARD	MATNR: Material Number WERKS: Plant LGORT: Storage Location	Storage Location Data
MAPE	MATNR: Material Number WERKS: Plant GZOLX: Preference zone	Export Control File
MBEW	MATNR: Material Number BWKEY: Valuation Area BWTAR: Valuation Type	Material Valuation
MBEWH	MATNR: Material Number BWKEY: Valuation Area BWTAR: Valuation Type LFGJA: Current Year LFMON: Current period	Material Valuation History
MVKE	MATNR: Material Number VKORG: Sales Organization VTWEG: Distribution Channel	Sales Data
MARM	MATNR: Material Number MEINH: Unit of Measure	Units of Measure
MEAN	MATNR: Material Number MEINH: Unit of Measure LFNUM: Consecutive Number	EANs
MLEA	MATNR: Material Number MEINH: Unit of Measure LIFNR: Vendor number LFNUM: Consecutive Number	Vendor-Specific EANs
MAMT	MATNR: Material Number MEINH: Unit of Measure LFDNR: Consecutive Number SPRAS: Language MTXID: Short Text ID	Texts per UoM and Text ID

MLGN	MATNR: Material Number LGNUM: Warehouse Number	Data per Warehouse Number
MLGT	MATNR: Material Number LGNUM: Warehouse Number LGTY: Storage Type	Data per Storage Type
MLAN	MATNR: Material Number ALAND: Country	Tax Classification
MAEX	MATNR: Material Number ALAND: Country GEGRU: Legal regulation	Legal Control
MAPR	MATNR: Material Number WERKS: Plant	Index for Forecast
MVER	MATNR: Material Number WERKS: Plant GJAHR: Fiscal Year PERKZ: Period Indicator ZAHRL: Follow-on Records	Material Consumption
MDMA	MATNR: Material Number BERID: MRP Area	MRP Area

Related Content

Related Documents

Related SAP Notes/KBAs

Material Master Number Ranges

Purpose

This content was written to explain Material Master number ranges

Overview

Number ranges are a generic functionality in SAP which is used by material master. For every material that your company uses you must create a material master record for it. This record is uniquely identified by a material number. You can assign mnemonic keys or nonmnemonic keys as material numbers depending on the method your company prefers. For this reason you have the following types of number assignment in the R/3 System:

1) External number assignment:

If your company uses mnemonic keys (normally alphanumeric) you enter the character string you want to use as the material number. Some characters cannot be used in external material numbers as they are used as wildcards. Forbidden characters are:

- Ampersand (&)
- Asterisk (*)
- Comma (,)

The check for these characters is done by function module CHECK_STRING_SPEC_CHARACTERS. If you want to restrict the

allowed characters for a material number refer to SAP note [1474828](#). (Check for allowed characters in the material number).

Sometimes a material number is created with a character which cannot be displayed in the current font, codepage or cannot be displayed at all as it is a control character. These characters will then be displayed as a '#', without being a real hash sign. The effect of this will be that the material will not be found if you type the material number (including the hash) by keyboard. However it will be found when entering the material number (including the 'indisplable' character) by copy and paste, or by range search.

To analyse this you can start the debugger and check how MARA-MATNR is displayed in hexadecimal representation. A real hash sign '#' will be displayed as '0023'. Our recommendation for such cases is to archive the material and to then re create it.

2) Internal number assignment:

If your company uses nonmnemonic keys you do not enter a material number when creating the material master record. Instead the system assigns a consecutive number to the material. This number is visible when you maintain the material master record.

You can define the required type of number assignment in Customizing for the Material Master in Define Number Ranges. After defining a number range you can flag it if external number assignment is required as an external number range. You then assign it to one or more material types. As a result when creating a material master record the type of number assignment allowed depends on the material type chosen. In the standard R/3 System one external number range and one internal number range can be defined for each material type or group of material types.

SAP recommends using internal number assignment. Since you can use match codes to select materials according to any desired criteria. It is not normally necessary to use a mnemonic external number. However external number assignment can be advantageous if you want to use your materials not only in the SAP R/3 System. Here too for example the field old material number and a relevant matchcode is available in the standard SAP R/3 System.

For Internal number assignment every internal number range get's a counter indicating the last issued material number. The next number to be issued is this counter plus one. The material number is issued not when storing the material but already when starting transaction MM01. Thus everytime the creation of a material is initiated but the material not stored a gap in the material numbers in table MARA is generated. The numbers are issued in function module NUMBER_GET_NEXT. If you need your own logic here, BADI_NUMBER_GET_NEXT should help.

Buffering An SAP system usually has one central database but several servers sharing the processing workload. Without buffering everytime a user working on the server wants to create a material the server would have to ask the database for the actual number to name the material. This is simple but very time-consuming. With buffering, when a server needs a number in order to create a material it will get not only one number but a whole quota say 1000 (this can be customized) and use this for the next materials to create till it is empty then it asks for another quota. This is much faster for you need only one database access instead of many. In case of buffering the actual value which you see in table MMNR is not the value of the latest issued material number as for the non-buffered case but the highest number assigned in a quota to any of the servers.

One example to explain this: Lets imagine a system with three servers A, B and C, the quotas are customized with 100 and the numbers till 4200 are already completely used.

- Server A got a quota from 4201 till 4300, and already used till 4215
- Server B got a quota from 4301 till 4400, and already used till 4377
- Server C got a quota from 4401 till 4500, and already used till 4429

Then if you create a new material the number will be assigned as 4216, 4378 or 4430, depending on which server you are working. But in transaction MMNR the actual value will be displayed as 4500. In case of problems with the buffered number range assignment it might be necessary to reset the buffer. This can be done by transaction SM56 by pressing 'Number range Buffer - Reset', entering the client and the object MATERIALNR, and executing. By this, the buffer on the respective server is reset. It might be necessary to reproduce this procedure on each server. Further information about buffering is available in SAP note [62077](#) (Info: Internal number assignment is not continuous)

Customizing material number ranges

The transaction to customize number range objects is SNRO. This transaction also provides a generic access to the respective number range. Material number ranges are maintained specific to the material type.

Material number ranges are maintained in customizing transaction MMNR. The respective material type first is assigned to a group. Afterwards a maximum of one external and one internal number range can be assigned to a group. These assignments are stored in table NRIV with the object 'MATERIALNR'.

In the customizing of the material types, transaction OMS2 there is a flag for switching off the check for the external number range. In this case, the number ranges can be checked during the creation of a material by user exit MGA00002 with its components EXIT_SAPLMG02_001 (Customer Exit: Internal Material Number Assignment) and EXIT_SAPLMG02_002 (Customer Exit: External Material Number Assignment). IS-Retail Customers use user exit MGW00002.

Related Content

Related Documents

Related SAP Notes/KBAs

[1474828](#) (Check for allowed characters in the material number).

[72286](#) - Lexicographical <-> non-lexicographical material no.

[62077](#) (Info: Internal number assignment is not continuous)

Material Master transactions

Purpose

This content was written to list Material Master's transactions

	Transaction	Description
Immediate Maintenance	MM01	Create material
	MM02	Change material
	MM03	Display material
	MM04	Display material change documents
	MM06	Flag material for deletion
	MMAM	Change Material Type
Scheduled Maintenance	MM11	Schedule the creation of a material
	MM12	Schedule the change of material
	MM13	Activate scheduled changes for a key date
	MM16	Schedule the deletion of a material
Mass Maintenance	MM17	Mass Change
Lists and overviews	MM50	MaList extendable materials
	MM60	Material list

	MMBE	Stock overview
Period Closing	MMPV	Close period
	MMPI	Initialize Period
Archiving	MM70	Sel. Materials Flagged for Deletion
	MM71	Reorganize Materials
	MM72	Display Archive of Materials
	MM73	Special Stocks: Preparation
	MM74	Archive Special Stocks
	MM75	Display Archive of Special Stocks
	SARA	With object MM_MATNR: Archive materials
	SARA	With object MM_SPSTOCK: Archive batches and special stocks
	SARI	Archive Information System
Logs	MM90	Analyze ALE Appl. Log for Material Master
	MM91	Delete ALE Appl. Log for Material Master
General customizing	OMS2	Material type
	OMCE	Define batch level
	OMSL	Material number format
	OX14	Define valuation level
	OVK1	Tax Categories by Country
	OMSF	Define Material Groups
	OMS4	Define Material Statuses
	OMT4	Attributes of System Messages
	OMSY	Maintain Company Codes for Materials Management
Field selection customizing	OMSR	Field selection
	OMT0	Make global settings (general control data)
Screen sequence customizing	OMT3	Screen sequence customizing
	OMT3B	Define Structure of Data Screens for Each Screen Sequence
	OMT3Z	Assign secondary screens
	OMT3R	Maintain Order of Main and Additional Screens
	OMT3E	Assign Screen Sequences to User/Material Type/Transaction/Industry Sector
Number range customizing	SNR0	With object 'MATERIALNR': Number range object maintenance

	MMNR	Material number number range customizing
International Article Numbers (EANs)	W4EN	Maintain Number Ranges for GTIN (EAN)
	W4ES	Maintain Number Range Categories for GTIN (EAN)
Units of Measurement	CUNI	Units of Measurement

Related Content

Related Documents

Related SAP Notes/KBAs

The role of history tables- period closing example

The role of history tables- period closing example

The purpose of this page is to explain how the history tables are written during period closing (transaction MMPV) with help of a small example.

The golden rules

1. History data will be created only if there was at least one material movement in the period before closing it.
2. There is at the maximum one entry per period in the history tables.

Example

Material 'A' has a stock of 10 pieces at the end of period 01.

1. In period 02 a goods receipt of 5 pieces occurs.
Now an entry is added in the history table for period 01 with a stock of 10 pieces.
Simultaneously, the current stock is increased to 15 pieces and the 'Current Period' field (LFMON) is set to 02.
2. In period 02, an additional goods receipt occurs with 2 pieces.
The history table is not affected by this operation since an entry already exists for period 01. However, the current stock is increased to 17 pieces.
3. In period 04, a goods issue of 4 pieces occurs.
Now the system adds an entry for period 03 with a stock of 17 pieces in the history table.
Simultaneously, the current stock is decreased to 13 pieces and the 'Current Period' field (LFMON) is changed from 02 to 04.
Since no goods movement has occurred in period 03, there is no entry for period 02 in the history table.

MARV-LFMON	01	02	03	04
MBEW-LFMON	unknown, because no information is populated when was the last goods movement before period 01	set to 02, because there was a goods movement in the period 02	still 02, because no goods movement occurred in period 03	04
MBEWH-LFMON	unknown, because no information is populated when was the last goods movement before period 01	set to 01, when goods movement in period 02 occurs		set to 03, when goods movement in period 04 occurs

MBEW-LBKUM	10	15+2	17	13
MBEWH-LBKUM	set to 10, when goods movement in period 02 occurs	no new entry created, hence in period 03 no movements occurred	set to 17, when goods movement in period 04 occurs	

MARV-LFMON: number of the period according to the customizing settings in transactions OB29 and OMSY. This value will be always shifted when running the period closer (transaction MMPV, report RMMMPEI).

LFMON in stock or valuation related tables: Period of the last goods movement, invoice verification, price change, or other function that has resulted in period-specific information being updated.

LBKUM: Total Valuated Stock

Related Content

Related Documents

Related SAP Notes/KBAs

For a detailed technical explanation please check the SAP note [193554](#).

Working with MRP Profile, MRP Type

Purpose

The purpose of this page is to show how changes in MRP Profile work.

Overview

Changes in MRP Profile, changing MRP Type.

MRP Profile

This is the **main body** of your WIKI Page where you talk through the system logic or behavior you are explaining. Use pictures to emphasize the points you are trying to make or to illustrate the transactions/selection screens/reports etc.

Message number MM306

Same as the first section, this is also part of the **main body** of the WIKI. You can add as many sections as you require to make the information you are explaining as clear and simple to follow as possible.

"MRP profile XY changed; background job has been scheduled".

Same as the first section, this is also part of the **main body** of the WIKI. You can add as many sections as you require to make the information you are explaining as clear and simple to follow as possible.

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Insert SAP Help links or other WIKI content link.
Please hyperlink the title of the related document

Example: [Explanation about how to create a WIKI page](#)

Related SAP Notes/KBAs

Insert links to any related SAP Notes/KBAs that support your topic or are related. Please hyperlink **ONLY** SAP Note or KBA number.

Example:

SAP Note [83020](#): What is consulting, what is support

SAP KBA [12345](#): This is an example KBA link

(Use the following Hyperlink with SAP Note/KBA number at end: <https://service.sap.com/sap/support/notes/123456>)

Use this structure to help you compose your contributions for WIKI and at the same time will ensure spelling and grammar.